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Optimization of tapered semiconductor optical amplifiers for picosecond pulse amplification

E. Gehrig[a)] and O. Hess
Advanced Technology Institute, School of Electronics and Physical Sciences, University of Surrey, Guildford, Surrey GU2 7XH, United Kingdom

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We present results of a fundamental device simulation study aimed at an optimization of high-power tapered semiconductor optical amplifiers for picosecond pulse amplification. Thereby, the microscopic light-matter coupling is described on the basis of a spatially resolved Maxwell–Bloch–Langevin description that takes into account many-body-carrier interactions, energy transfer between the carrier and phonon systems and, in particular, the spatiotemporal interplay of stimulated and amplified spontaneous emission and the noise caused by spontaneous emission. Extensive simulation runs reveal the microscopic physical processes which are responsible for the beam quality of the amplified picosecond pulses and allow us to give concrete suggestions for the optimization of the amplifier structure and geometry. We discuss the influence of a curved waveguide or spatially patterned current contact, and identify the length of the unpumped area as the most critical parameter to be carefully optimized. © *2005 American Institute of Physics*.
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The need for high output power and good spatial and spectral purity, often required by applications in, e.g., ultrafast nonlinear optics or communication systems, has lead in recent years to device designs for high-power high-brightness semicondcutor laser sources[1,2] and to the realization of new laser amplifiers with improved beam quality.[3–7] The laser amplifier system consisting of a a laser pulse source and a semiconductor amplifier is a very commonly used configuration since it allows the separate optimization of device properties responsible for pulse properties and output power. The spatial and spectral properties of a pulsed signal that has been amplified in a high-power semiconductor laser thereby strongly depend on both the geometry of the laser (in particular the waveguide geometry) and the specific realization of the current injection via an electronic contact. More recent investigations[8] on tapered laser amplifiers have revealed the influence of the tapered angle and the length of the waveguide section on the beam quality and power of the amplified signal. It could be shown that a tapered angle that has been adjusted to the diffraction angle of the injected Gaussian beam and a waveguide length of a few 100 μm provide good beam quality and high small signal gain with only small contributions of amplified spontaneous emission (ASE) in the emitted radiation.

While previous studies have mainly addressed the amplification of continuous-wave (cw) optical signals, in this letter, we focus on an investigation of the design of the waveguide section and the electronic contact of tapered laser amplifiers (InGaAs, 2–3 mm length) that should be optimized for picosecond pulse amplification. In particular, we will analyze the influence of these design parameters with respect to the spatiotemporal light field dynamics and quality of the picosecond pulses. Table I summarizes the fundamental material and device parameters used in the modeling. More details on the background and on their significance can be found in Ref. 9. In our simulations, the microscopic light-matter coupling is described on the basis of a spatially resolved Maxwell–Bloch–Langevin description[9] taking into account many-body-carrier interactions, energy transfer between the carrier and phonon systems and, in particular, the spatiotemporal interplay of stimulated and ASE and the noise caused by spontaneous emission. The scattering rates are microscopically derived and calculated in dependence on carrier density and wavenumber.[9] Figure 1 shows a lateral cut of the intensity distributions at the output facet [Figs. 1(a) and 1(b)] and in the entrance region [Figs. 1(c) and 1(d)] of the taper (approximately 20 μm behind the waveguide). The length of the waveguide is 770 μm, the total length of the amplifier is 2770 μm). Figures 1(a) and 1(c) show simulation results for a tapered amplifier with a linear waveguide, Figs. 1(b) and 1(d) refer to a device with a curved waveguide section. Please note that the full spatiotemporal carrier and

TABLE I. Fundamental material and device parameters of the InGaAs tapered lasers.

Material	Parameter
Cavity length	2770 μm
Length of the waveguide	770 μm
Width at the entrance facet	5 μm
Width at the output facet	200 μm
Thickness of active layer	0.15 μm
Refractive index of active layer	3.35
Laser wavelength	925 nm
Front facet mirror reflectivity	10^{-2}
Back facet mirror reflectivity	10^{-2}
Nonrad. recomb. time	5 ns
Relaxation times of electrons and holes	50–200 fs
Dipole dephasing time	50–200 fs
Effective electron mass	0.065–0.068m_0
Effective hole mass	0.24–0.38m_0
m_0	9.1093879×10^{-31} kg
Energy gap (at T=0 K)	1.35–1.529 eV
Injection efficiency	0.5

[a)]Electronic mail: e.gehrig@surrey.ac.uk

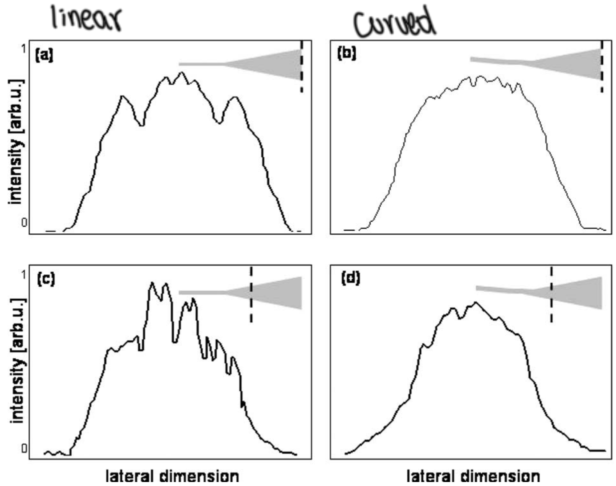

FIG. 1. Intensity distributions (temporally averaged) at the output facet (upper row) as well as in the entrance region of the taper (bottom row) with linear (left) and curved (right) waveguide.

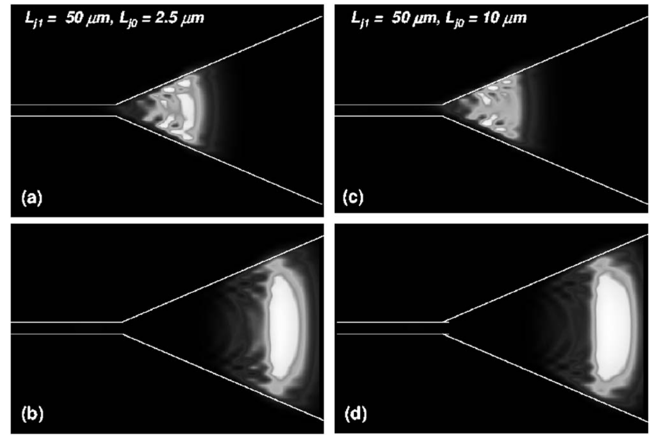

FIG. 2. Propagation of a picosecond pulse in a tapered amplifier with a length of the unpumped area of L_{j0}=2.5 μm (a,b) and L_{j0}=10 μm (c,d).

light field dynamics have been taken into account in the calculation. Indeed, it is the dynamic spatially dependent interplay of carrier diffusion, light diffraction, and microscopic carrier excitation and relaxation that affect the spatial and spectral properties of the light pulse during its propagation. In the calculations, the injection current was chosen such that the carrier inversion within the amplifier is signficantly above the threshold value ($N \approx 3N_{\text{thr}}$), and the input power is $1.1P_{\text{in}}^{\text{sat}}$—where $P_{\text{in}}^{\text{sat}}$ denotes the input power required to reduce the inversion to transparency when reaching the output facet. The material reflectivity of the facets has been set to a value of 10^{-2}. In the linear geometry [Figs. 1(a) and 1(c)], the dynamic interaction of counterpropagating waves leads to characteristic transverse modulations in the entrance region of the taper [Fig. 1(a)] that are via the dynamic light field propagation transferred to respective filament structures in the near field [Fig. 1(b)]. The nonlinear waveguide geometry reduces and suppresses counterpropagation effects leading to a homogeneous light field distribution near the waveguide-amplifier connection [Fig. 1(b)] and an almost Gaussian-shaped near field [Fig. 1(d)]. Generally, the interaction of counterpropagating waves strongly depends on the quality of the antireflexion coating. For cw beams, it has been shown[10] that the facet reflectivity of tapered amplifiers with linear geometry should be less than 10^{-4}. However, as can be seen in Fig. 1, a laser-amplifier design with curved waveguide section modifies these requirements: The nonlinear waveguide section leads to an efficient coupling of the injected light field into the amplifying region and to a significant reduction of counterpropagation effects. In a tapered amplifier with a curved waveguide section, the "effective reflectivity" defined by both the reflectivity of the facets (including coating) and the phase front of the counterpropagating light wave is thus reduced by approximately two orders of magnitude.

A further increase in quality and power of the optical signal (cw beam or pulse alike) can be obtained with an appropriate design of the electronic contact. As expected, the carrier injection via the contact leads to a characteristic carrier distribution within the active area of the laser amplifier. However, the geometric design of the contact plays a significant role for the resulting spatiotemporally varying light field distribution and filament formation. A typical method to control the carrier-light field dynamics is to use a spatially modulated contact. The suitable choice of, e.g., the length and periodicity of the modulated section, however, often remains an open question. A systematic variation of the shape and periodicity of the current contacts should then allow us to determine the influence of a spatially varying carrier injection. Specifically, we consider a laser amplifier in which the electronic contact in the entrance region of the taper is periodically modulated in propagation direction. Variable parameters are the length of the modulated section (L_{mod}), the length of a single contact region (L_{j1}), as well as the length of the umpumped regions (L_{j0}). One might immediately wonder, if and how the structuring of the contact may influence the spatiotemporal dynamics and thus keep filament formation at a low level. In order to answer this question, we have systematically varied the characteristic lengths L_{mod}, L_{j1}, and L_{j0}. Figure 2 shows calculated intensity snapshots pertaining to a tapered amplifier ($N \approx 3N_{\text{thr}}$, length of the modulated section L_{mod}=600 μm) during the propagation of a picosecond pulse. In this particular example, the length of the umpumped area is L_{j0}=2.5 μm (left) and L_{j0}=10 μm (right), respectively, and the input power is $0.8P_{\text{in}}^{\text{sat}}$. In principle, the inhomogeneous contact design keeps the injection current density in the entrance region at a low level and thus reduces lateral structuring and filament formation. The spatiotemporal dynamics near the waveguide-taper transition via propagation and the dynamic coupling of transverse with longitudinal degrees of freedom is also transferred to the propagation direction leading to good pulse-signal properties near the output facet. However, it is important to note that the laser-internal light-matter interaction and the respective characteristic interaction length (e.g., carrier diffusion, light diffraction) give upper and lower limits to the contact design, respectively. This is of particular importance for the length of the umpumped area: If this length significantly exceeds the laser-internal interaction length (which typically lies in the μm regime), a detoriation instead of an improvement of the beam quality will arise. Our simulations demonstrate that large umpumped regions lead to an inhomogeneous amplification within the pulse shape [see Fig. 2(c)]. This results in a loss of phase information and coherence. The pulse breaks up and is significantly deformed [Fig. 2(d)]. In addition, a significant detoriation of the gain can be ovserved: Fig. 3(a) shows (for the same current and input power) the spatiotemporally integrated values of the gain as calculated within the frame of the complete set of spatiotemporally resolved

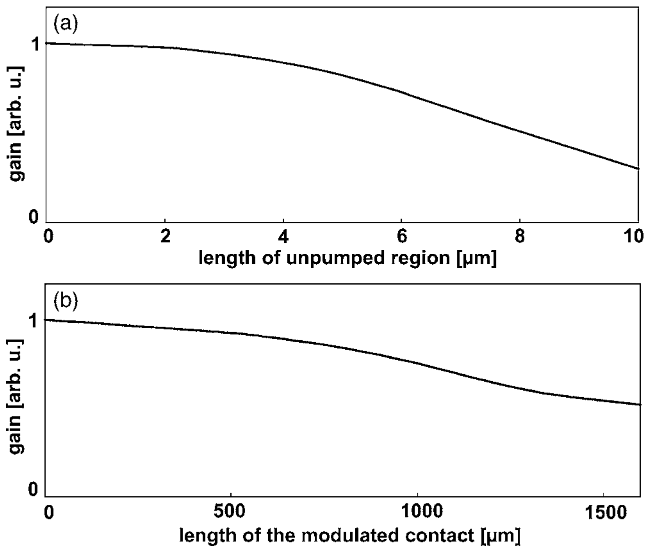

FIG. 3. Dependence of the overall gain on the length of the unpumped region (a) and on the total length of the modulated area (b) in an amplifier with a total length of 2770 μm.

Maxwell–Bloch equations. If the length of the unpumped region is significantly larger than approximately 5 μm, the induced phase changes lead to a reduction of the over-all gain. This effect is even more pronounced if the input power is very low (i.e., in the low signal regime): In this case, the pulse amplitude is not sufficiently high to sustain internal coherence during the passage of the umpumped regions so that the shape and duration of the injected pulse is not maintained. The laser-internal interaction lengths given by diffusion, diffraction, and scattering (typically in the μm regime) consequently represent and define the maximum length of the umpumped region.

To gain further insight in the influence of the contact modulation on the amplification, Fig. 3(b) displays the dependence of the integrated total gain on the length of the modulated section in the taper. In structures with a long modulated section, our simulations reveal a noticable influence of the curvature of the phase front defined by the propagation of the injected Gaussian beam (i.e., beam waist and wavelength-dependent diffraction): With increasing propagation length, the curvature of the phase front of the light field deviates to a stronger degree from an even phase front. As a consequence, discontinuities arise where the propagating light fields encounter the perpendicularly arranged sequence of pumped and unpumped regions. If the length of the modulated area extends too far in the taper (more than, e.g., 1 mm) this "mismatch" between phase curvature and contact sections leads to a strong reduction in the overall amplification [see Fig. 3(b)].

The results of our parameter variation can be summarized as follows. A curved waveguide significantly reduces counterpropagation effects. A spatially structured contact section allows, for suitable design, filament supression and consequently improves the quality of an amplified picosecond pulse. However, to prevent a transverse deterioation of the signal associated with the curved phase front of the propagating Gaussian beam, the modulated contact should not cover the whole taper but be restricted to an area immediately behind the waveguide. Finally, the most critical parameter is represented by the length of the unpumped area. For low input powers, the value of this parameter should not significantly exceed the μm regime in order to sustain the coherence properties of the light field during propagation.

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